A3-PS017-TC202

Tail-Latency Prediction for URLLC-Like Traffic In Simulation

Ravi Shankar Thota
ravishankarthota32@gmail.com

AIORI Hackathon 3

Problem Description

- Network congestion in 5G/6G networks can cause some data packets to experience unusually high delays, Known as tail latency.
- Average latency alone cannot identify rare delays, So P95 and P99 metrics are used to analyze network performance.
- Machine Learning is used to predict P99 tail latency based on simulated network conditions such as traffic load, queue length, and congestion.
- The project aims to simulate URLLC-like network traffic under different conditions, including traffic load, packet size, number of users, queue length, and network congestion.
- A Streamlit dashboard displays latency predictions and identifies potential high-latency risks to support reliable, low-latency communication.

Background

- 6G and future networks require ultra-low latency and high reliability to support applications such as autonomous vehicles, industrial automation, and real-time communication.
- Network congestion, increasing traffic load, and long queues can cause some requests to experience unusually high delays, known as tail latency.
- Average latency alone may not reveal these rare delays; therefore, P95 and P99 latency metrics are important for evaluating network performance.
- This project uses network simulation and Machine Learning to predict P99 tail latency and identify potential high-latency risks, helping improve the reliability of future networks.

Literature & RFC Review

Existing research and network standards highlight the importance of low latency and high reliability in 5G/6G networks, particularly for Ultra-Reliable Low-Latency Communication (URLLC).

- Tail-Latency Analysis: P95 and P99 percentiles are used to measure delays experienced by the slowest network requests.
- Network Simulation: Simulated traffic helps evaluate latency under different traffic loads, queue lengths, and congestion conditions.
- Machine Learning: Regression models can predict latency patterns using network performance parameters.
- Our Approach: We combine network simulation, P99 latency prediction, and a Streamlit dashboard to identify potential high-latency risks in future networks.

Proposed Solution

Our proposed solution uses network simulation and Machine Learning to predict tail latency in URLLC-like traffic.

- Generate simulated network traffic under different conditions, including traffic load, number of users, packet size, queue length, and congestion.
- Calculate network latency and analyze P95 and P99 metrics to identify high-delay conditions.
- Train a Random Forest Regression model to predict P99 tail latency using simulated network parameters.
- Develop a Streamlit dashboard to display predicted latency, performance graphs, and high-latency risk alerts.

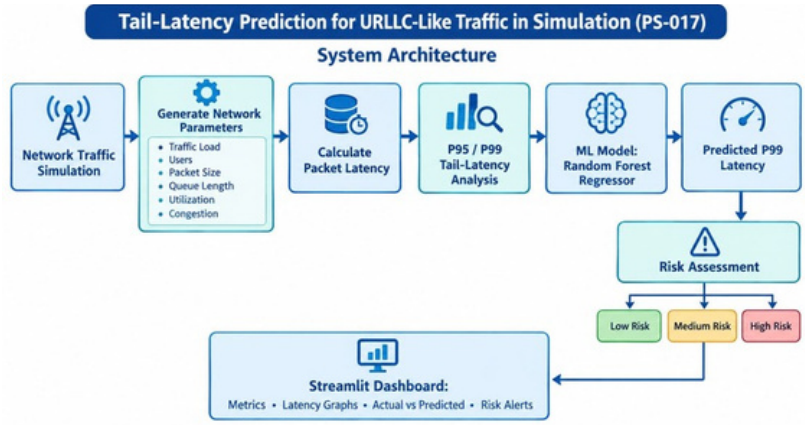
The system aims to identify potential latency problems early and support reliable, low-latency communication in future 6G networks.

Optimization Proposed

- Predict P99 tail latency using Machine Learning under different simulated network conditions.
- Analyze traffic load, queue length, network utilization, and congestion to identify potential high-latency risks.
- Compare latency performance under low, medium, and high traffic scenarios.
- Use P95 and P99 metrics to evaluate network performance beyond average latency.
- Develop an interactive Streamlit dashboard to display predictions, graphs, and high-latency risk alerts.
- Provide early warnings about potential latency issues to support reliable, low-latency communication in future 6G networks.

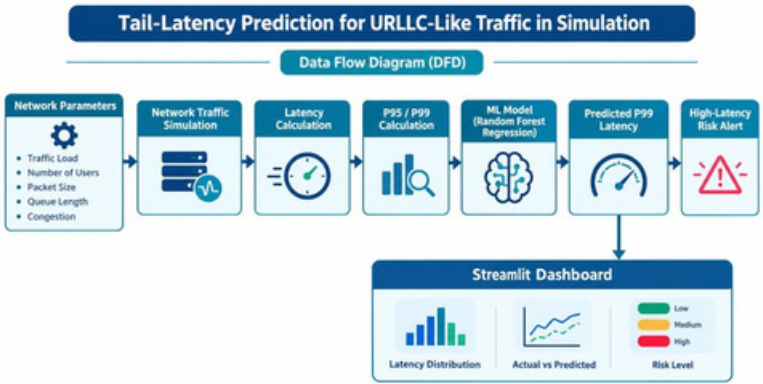
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

- Phase 1: Generate a simulated network traffic dataset with different network conditions.
- Phase 2: Calculate latency metrics, including P95 and P99, and analyze tail-latency behavior.
- Phase 3: Train and evaluate a Machine Learning model to predict P99 tail latency.
- Phase 4: Develop a Streamlit dashboard to visualize latency predictions, performance graphs, and risk alerts.
- Phase 5: Test the complete system under low, medium, and high traffic scenarios and present the results.

Bibliography I

[1] A. Sriraman et al., “Deconstructing the Tail at Scale Effect Across Network Protocols,” 2016.

[2] J. Li, N. K. Sharma, D. R. K. Ports, and S. D. Gribble, “Tales of the Tail: Hardware, OS, and Application-level Sources of Tail Latency,” ACM Symposium on Cloud Computing, 2014.

[3] 3GPP, “Study on Enhancement of Ultra-Reliable Low-Latency Communication (URLLC) Support in the 5G Core Network,” Technical Report 23.725.

Thank You for Your Attention!